PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Consolidation chemotherapy in postmolar low-risk gestational trophoblastic neoplasia: a systematic review protocol
AUTHORS	Branco-Silva, Mariza; Maesta, Izildinha; Elias, Kevin; Berkowitz, Ross; Abbade, Joelcio; Horowitz, Neil

VERSION 1 – REVIEW

REVIEWER	Winter, Matthew Weston Park Cancer Centre
REVIEW RETURNED	31-Dec-2021

GENERAL COMMENTS	Thank you for asking me to review this proposed systematic review protocol from internationally renowned GTN specialists evaluating whether consolidation chemotherapy is required for all patients with post-molar GTN after hCG normalisation with 1st-line single agent chemotherapy. Although the cure rate for low risk GTN should approximate 100%, this is an important question aiming to review the role of consolidation chemotherapy treatment and whether de-escalation (in terms of need for consolidation treatment and duration/number of cycles), without impacting on recurrence/ relapse, can be considered, and in which patients this strategy may be considered to reduce burden of treatment. The use of consolidation chemotherapy following normalisation of hCG in low-risk GTN is variable throughout the world with no agreed consensus approach. For example the European (EOTTD, 2020) guidelines recommend that once hCG has normalised patients are treated for a minimum of four consecutive weeks with consolidation according to local GTD centre advice. Within Europe this may be 2-3 consolidation cycles with 3 cycles (6 weeks) given in the United Kingdom. The protocol is very clear. There are some limitations in that the number of randomised trials available will be very small. The study team plan to include non-randomised data as outlined in the protocol which is appropriate given the relative rarity of GTN. I note the study timeframe is up to 12 months of follow-up after hCG normalisation with first-line chemotherapy. The risk of recurrence following treatment for low-risk GTN approximates 5%, and of these the majority do occur within the first 12 months (approx. 73%). However there are some patients that relapse after 12 months. It would be useful to know how the investigators will account for these later, albeit less common, recurrences.

REVIEWER	Joneborg, Ulrika Karolinska University Hospital, Department of Pelvic Cancer/Gynecologic Oncology
REVIEW RETURNED	05-Jan-2022

GENERAL COMMENTS	Thank you for allowing me to review this systematic review protocol. The research question is very interesting and the results could lead to a more individualized treatment of women with post-molar low-risk GTN. I have some minor comments to three of the questions in the checklist. 3. The study design is appropriate for this research question, however, since the number of RCTs, as the authors have concluded, is expected to be very low, if any, the authors might face difficulties in drawing a reliable conclusion. The proposed systematic review is however currently the best way of answering this research question, but a multi-center RCT with enough power is desirable. 8. The authors should use the same reference system throughout the text. Most references are noted as numbers and brackets, but a few as author names and publication year in the text, ie page 5 lines 88, 89, 95. 15. The authors should use a uniform term for gestational (gestation) trophoblastic neoplasia, and change gestation to gestational on page 3 line 53. Also, numbers in the text should be written in a uniform way with letters, ie not two and 2 or three and 3, see page 5 lines 107 and 108 and page 6 lines 110, 116, 120.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Thank you for asking me to review this proposed systematic review protocol from internationally renowned GTN specialists evaluating whether consolidation chemotherapy is required for all patients with post-molar GTN after hCG normalisation with 1st-line single agent chemotherapy.

Response: Thank you very much for your time and consideration.

Although the cure rate for low risk GTN should approximate 100%, this is an important question aiming to review the role of consolidation chemotherapy treatment and whether de-escalation (in terms of need for consolidation treatment and duration/number of cycles), without impacting on recurrence/ relapse, can be considered, and in which patients this strategy may be considered to reduce burden of treatment.

Response: We completely agree that it is very important to investigate the prognostic factors of recurrence of low-risk postmolar gestational trophoblastic neoplasia (GTN) so that we can consider consolidation chemotherapy necessary only for these patients with low-risk postmolar GTN.

The use of consolidation chemotherapy following normalisation of hCG in low-risk GTN is variable throughout the world with no agreed consensus approach. For example the European (EOTTD, 2020) guidelines recommend that once hCG has normalised patients are treated for a minimum of four consecutive weeks with consolidation according to local GTD centre advice. Within Europe this may be 2-3 consolidation cycles with 3 cycles (6 weeks) given in the United Kingdom.

Response: Indeed, no agreed consensus approach has been reached about the use of consolidation chemotherapy following normalization of hCG in low-risk GTN. In South America, for example, consolidation chemotherapy has been administered using the last effective regimen as one cycle to all patients in Argentina (Carlos G Durand Hospital Trophoblastic Disease Center, Buenos Aires), while in Brazil (Botucatu Trophoblastic Center) and Colombia (Oncólogos del Occidente S.A. Manizales, Caldas) two consolidations cycles have been given for stage II or III cases (unpublished data). Moreover, according to the Society of Gynecologic Oncology (Horowitz et al., 2021) evidenced-based review and recommendation, the evidence regarding three consolidation cycles is Level II-2.

The protocol is very clear. There are some limitations in that the number of randomised trials available will be very small. The study team plan to include non-randomised data as outlined in the protocol which is appropriate given the relative rarity of GTN.

Response: Thank you for your comment. Indeed, the number of randomized controlled trials found, if any, is expected to be very small as this type of study usually requires a large sample size, not easily available in the context of rare cancers such as GTN. However, we believe that including non-randomized studies will help obtain maximum information from published data and identify areas for improvement. Moreover, a rigorous risk of bias assessment and grading of the quality of the evidence will add confidence in the study's results.

I note the study timeframe is up to 12 months of follow-up after hCG normalisation with first-line chemotherapy. The risk of recurrence following treatment for low-risk GTN approximates 5%, and of these the majority do occur within the first 12 months (approx. 73%). However there are some patients that relapse after 12 months. It would be useful to know how the investigators will account for these later, albeit less common, recurrences.

Response: Thank you for your useful comment. We have rephrased the sentence to "at least 12 months of follow-up after hCG normalization with first-line chemotherapy".

We had chosen to use a 12-month time frame considering that most studies (Ngan et al., 2021, Horowitz et al., 2021, Ngan et al., 2018) recommend the surveillance of low-risk GTN relapse for at least 12 months and that patients should receive reliable contraception throughout the entire duration of follow-up. After this period, these women are free to get pregnant and a second tumor may be mistaken as recurrence.

However, as you pointed out, the largest study (Balachandran et al., 2020) on the duration of hCG surveillance following chemotherapy for GTN showed that the greatest risk of recurrence occurred in the first year after completing treatment measuring 72.7% (n= 112/154), while in the second year it is 12.3% (n=19/154).

Thus, we believe that by considering the first and second year after completing first-line chemotherapy we can measure 85% of all recurrence cases.

Reviewer: 2

Comments to the Author:

Thank you for allowing me to review this systematic review protocol. The research question is very interesting, and the results could lead to a more individualized treatment of women with post-molar low-risk GTN. I have some minor comments to three of the questions in the checklist.

Response: Thank you very much for your comments and suggestions.

3. The study design is appropriate for this research question, however, since the number of RCTs, as the authors have concluded, is expected to be very low, if any, the authors might face difficulties in drawing a reliable conclusion. The proposed systematic review is however currently the best way of answering this research question, but a multi-center RCT with enough power is desirable.

Response: We totally agree that a multi-center RCT is desirable, but, unfortunately, RCTs pose special barriers in the field of rare diseases such as GTN.

8. The authors should use the same reference system throughout the text. Most references are noted as numbers and brackets, but a few as author names and publication year in the text, ie page 5 lines 88, 89, 95.

Response: Thank you for catching these errors, which we have now corrected.

15. The authors should use a uniform term for gestational (gestation) trophoblastic neoplasia, and change gestation to gestational on page 3 line 53. Also, numbers in the text should be written in a uniform way with letters, ie not two and 2 or three and 3, see page 5 lines 107 and 108 and page 6 lines 110, 116, 120.

Response: Thank you again for catching these errors, which we have now corrected.